# Neutron Kinetics with Delayed Neutrons

NE 630 | Nuclear Reactor Theory

**27** LESSON

**Read:** FNRP 5.4.

Students will be able to write down, solve, and explain each term of a first-order, ordinary differential equation that models the populations of neutrons and delayed-neutron precursors as functions of time.

## 1 Review: Subcritical Multiplication

Consider

$$\frac{dn}{dt} = \overbrace{[(k_\infty - 1)/l_\infty]}^{\rho_\infty/\Lambda_\infty} n(t) + S_0 \,. \qquad \text{(FNRP 5.22)}$$

What happens to $n(t)$ as $k_\infty \to 1$?

## 2 Delayed Neutron Precursors

Each fission produces two, neutron-rich fission fragments.

Some of these fragments produce radioactive daughters that decay by neutron emission, and we call the initial nuclide a **delayed neutron precursor** or just **precursor**. The total fraction of all neutrons (eventually) produced by fission that are delayed is called $\beta$.

For thermal fission of $^{235}$U, about two-thirds of these delayed neutrons are produced by just 10 nuclides (out of hundreds).

| Precursor | $t_{1/2}$ (s) | %$\beta$ |
|---|---|---|
| $^{137}$I | 24.5 | 12.8 |
| $^{94}$Rb | 2.8 | 9.6 |
| $^{89}$Br | 4.4 | 9.3 |
| $^{90}$Br | 1.9 | 8.1 |
| $^{88}$Br | 16.3 | 6.7 |
| $^{85}$As | 2.1 | 5.4 |
| $^{138}$I | 6.5 | 4.6 |
| $^{139}$I | 2.3 | 4.1 |
| $^{95}$Rb | 0.4 | 3.9 |
| $^{87}$Br | 55.7 | 2.9 |
| **subtotal** | | **67.5** |

## 3 Precursors By Example

Among the most important precursors is $^{87}$Br, which leads to neutron emission via

$$^{236}\text{U}^* \xrightarrow[\gamma=0.01274]{\text{fission}} {}^{87}\text{Br} \xrightarrow[t_{1/2}=55.65\text{ s}]{\beta^-} {}^{87}\text{Kr}^* \xrightarrow[f=0.0260]{n} {}^{86}\text{Kr}\,,$$

and where the final decay of $^{87}$Kr$^*$ is essentially instantaneous.

As we did for $^{135}$Xe, we can explicitly track the concentration of $^{87}$Br in time for a *constant* flux by solving

$$\frac{dN_{\text{Br-87}}}{dt} = -\lambda_{\text{Br-87}} N_{\text{Br-87}}(t) + \gamma_{\text{Br-87}} \Sigma_f \phi \,. \qquad (27.1)$$

**Example**: What is the steady-state concentration of $^{87}$Br?

**Ans.** $N^\infty_{\text{Br-87}} = \frac{\gamma_{\text{Br-87}} \Sigma_f \phi}{\lambda_{\text{Br-87}}}$.

## 4 The Delayed Neutron Precursor Fraction

The total neutron emission rate density (in steady state!) is

$$\begin{aligned} R_n = \bar{\nu}\Sigma_f\phi &= \bar{\nu}_p \Sigma_f \phi + f^n_{\text{Br-87}} \lambda_{\text{Br-87}} N^\infty_{\text{Br-87}} \\ &= (\bar{\nu}_p + f^n_{\text{Br-87}}\gamma)\Sigma_f\phi \\ &= (\bar{\nu}_p + \bar{\nu}_d)\Sigma_f\phi \,, \end{aligned}$$

where $p$ and $d$ indicate *prompt* and *delayed*. Thus, the fraction of all neutrons emitted from an initial fission event that are delayed is

$$\frac{\bar{\nu}_d}{\bar{\nu}} = \frac{f^n_{\text{Br-87}}\gamma_{\text{Br-87}}}{\bar{\nu}} = \frac{f^n_{\text{Br-87}}\gamma_{\text{Br-87}}}{\bar{\nu}_p + f^n_{\text{Br-87}}\gamma_{\text{Br-87}}} \equiv \beta_{\text{Br-87}} \,.$$

**Example**: Compute $\beta_{\text{Br-87}}$ for $^{235}$U (for which $\bar{\nu} = 2.43$).

**Ans.** $\beta_{\text{Br-87}} = 0.0260(0.01274)/2.43 \approx 0.000136$.

## 5 The Delayed Neutron Precursor Concentration

With $\beta_{\text{Br-87}}$, we can modify the neutron kinetics equation (FNRP 5.22) with the equation for $N_{\text{Br-87}}$, or

$$\begin{aligned} \frac{dn}{dt} &= \left(\frac{\rho - \beta_{\text{Br-87}}}{\Lambda}\right) n(t) + f_{\text{Br-87}} \lambda_{\text{Br-87}} N_{\text{Br-87}}(t) + S(t) \\ \frac{dN_{\text{Br-87}}}{dt} &= -\lambda_{\text{Br-87}} N_{\text{Br-87}}(t) + \gamma_{\text{Br-87}} \Sigma_f \phi(t) \,. \end{aligned}$$

Although complete, we can simplify the notation a little bit by defining

$$C_{\text{Br-87}} = f_n N_{\text{Br-87}}$$

where $C_{\text{Br-87}}$ represents the concentration of only those $^{87}$Br nuclei that ultimately decay by neutron emission.

## 6 The Single-Precursor Equations

Rewriting the equations in terms of $C_{\text{Br-87}}$ leads to

$$\boxed{\frac{dn}{dt} = \left(\frac{\rho - \beta}{\Lambda}\right) n(t) + \lambda_{\text{Br-87}} C_{\text{Br-87}}(t) + S(t)} \,,$$

and

$$\boxed{\frac{dC_{\text{Br-87}}}{dt} = -\lambda_{\text{Br-87}} C_{\text{Br-87}}(t) + \beta_{\text{Br-87}} \bar{\nu} \Sigma_f \overbrace{\bar{\nu} n(t)}^{\phi(t)}} \,.$$

Though specific to $^{87}$Br, these equations have the proper form for single-precursor kinetics!

**Example**. Suppose the only precursor were Br-87. If $l = l_p = 10^{-5}$ s, what is the average lifetime of all neutrons? Assume that a delayed neutron's lifetime starts at the fission and ends with its death.

**Ans.** $l_d = l_p + 55.54/\ln(2)$ and $\bar{l} = (1-\beta)l_p + \beta l_d \approx 0.01$ s.

## 7 The Kinetics Equations with Multiple Precursors

Because there are hundreds of precursors, it a universal practice is to lump the precursors into groups defined by effective $\beta$'s and $\lambda$'s for each fissile nuclide of interest as in Table 5.1 from FNRP:

| Approximate Half-life (s) | Delayed Neutron Fraction $\beta_i$ | | |
|---|---|---|---|
| | $^{233}$U | $^{235}$U | $^{239}$Pu |
| 56 | 0.00023 | 0.00021 | 0.00007 |
| 23 | 0.00078 | 0.00142 | 0.00063 |
| 6.2 | 0.00064 | 0.00128 | 0.00044 |
| 2.3 | 0.00074 | 0.00257 | 0.00069 |
| 0.61 | 0.00014 | 0.00075 | 0.00018 |
| 0.23 | 0.00008 | 0.00027 | 0.00009 |
| **Total delayed fraction,** $\beta = \sum_{i=1}^{N} \beta_i$ | **0.00261** | **0.00650** | **0.00210** |
| **Total neutrons/fission,** $\bar{\nu} = \bar{\nu}_p + \bar{\nu}_d$ | **2.50** | **2.43** | **2.90** |

## 8 The Point Kinetics Equations (PKEs)

Using data from Table 5.1, the point kinetics equations are defined as

$$\boxed{\frac{dn}{dt} = \left(\frac{\rho(t) - \beta}{\Lambda}\right) n(t) + \sum_{i=1}^{N} \lambda_i C_i(t) + S(t)} \qquad \text{(FNRP 5.47)}$$

and

$$\frac{dC_i}{dt} = -\lambda_i C_i(t) + \beta_i \bar{\nu} \Sigma_f \bar{\nu} n(t) \quad i = 1, 2, \ldots, N$$

$$\boxed{\frac{dC_i}{dt} = -\lambda_i C_i(t) + \frac{\beta_i}{\Lambda} n(t) \quad i = 1, 2, \ldots, N}. \qquad \text{(FNRP 5.48)}$$

where $\Lambda = (\bar{\nu}\Sigma_f \bar{\nu})^{-1}$.

**Ponderable**: Above, $\rho(t)$ is a function of time. What real-world things could lead to a time-dependent reactivity?

## 9 Specifying Initial Conditions for PKEs

Of course, such sets of *first-order, ordinary differential equations* are not complete without appropriate **initial conditions**! The most common initial condition for reactor kinetics is to start from a critical, steady-state configuration at a known power.

**Example**. Suppose a reactor is operating at a constant power $P_0$ such that the neutron population is $n_0$.

- Determine an expression for the initial values $C_i(0)$.
- Using the precursor data from Table 5.1, $n_0 = 10^8$, and $\Lambda = 5 \cdot 10^{-5}$ s, determine each of the initial precursor concentrations.

**Ans.** From FNRP 5.48, set $dC_i/dt$ to zero and find $C_i(0) = (\beta_i n_0)/(\lambda_i \Lambda)$ for $i = 1, 2, \ldots, 6$. Then, substitute numerical data from Table 5.1 to find, e.g., $C_1(0) \approx 3.39 \cdot 10^{10}$ cm$^{-3}$. Quite often, $C_i(0) \gg n(0)$!

## 10 Numerical Solutions of PKEs

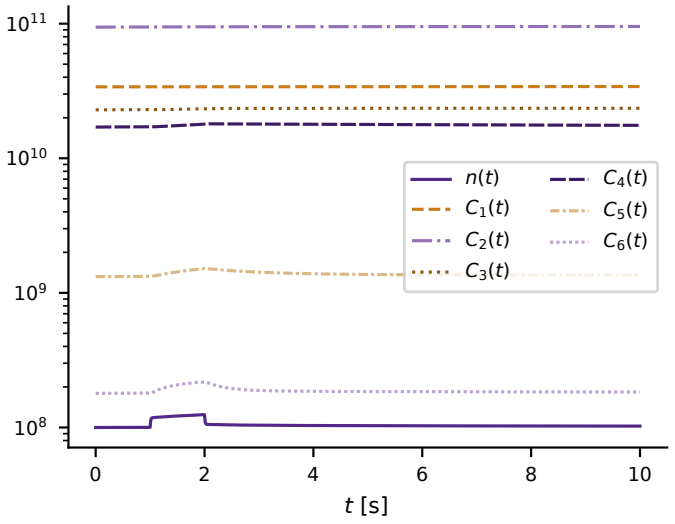
Most computer tools for solving systems of first-order equations require that the system be written in the following form:

$$\frac{d}{dt}\mathbf{y} = \mathbf{f}(\mathbf{y}(t), t)\,.$$

Equations (5.47) and (5.48) can be written as

$$\overbrace{\frac{d}{dt}\begin{bmatrix} n(t) \\ C_1(t) \\ \vdots \\ C_N(t) \end{bmatrix}}^{\mathbf{y}} = \overbrace{\begin{bmatrix} (\rho(t)-\beta)/\Lambda & \lambda_1 & \ldots & \lambda_N \\ \beta_1/\Lambda & -\lambda_1 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \beta_N/\Lambda & 0 & \ldots & -\lambda_N \end{bmatrix}}^{\mathbf{A}} \times \begin{bmatrix} n(t) \\ C_1(t) \\ \vdots \\ C_N(t) \end{bmatrix} + \overbrace{\begin{bmatrix} S(t) \\ 0 \\ \vdots \\ 0 \end{bmatrix}}^{\mathbf{b}}.$$

so we already have the form needed where $\mathbf{f}(\mathbf{y}, t) = \mathbf{A}\mathbf{y} + \mathbf{b}$. Example code based on SciPy's `odeint` is available on Canvas in the Jupyter notebook.

## 11 Numerical Solutions of PKEs

Using the precursor data from Table 5.1, $n_0 = 10^8$, and $\Lambda = 5 \cdot 10^{-5}$ s, $n(t)$ and $C_i(t)$ are shown below for a small step insertion of reactivity between 1 and 2 seconds. Example code is on Canvas.